

KABUL INTL, Afghanistan

WMO: 409480

Lat: 34.5552N

Lon: 69.2142E

Elev: 1791

StdP: 81.58

Time Zone: 4.50 (AFG)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.2	(c) -6.8	(d) -20.9	(e) 0.7	(f) 2.2	(g) -18.8	(h) 0.9	(i) 3.4	(j) 6.7	(k) 3.6	(l) 5.7	(m) 2.7	(n) 2.9	(o) 310	(p) 0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.2	(c) 35.2	(d) 14.6	(e) 34.1	(f) 14.3	(g) 33.1	(h) 14.2	(i) 19.7	(j) 27.0	(k) 18.9	(l) 26.2	(m) 18.0	(n) 25.5	(o) 4.4	(p) 350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	15.9	23.8	16.2	14.4	23.5	15.0	13.3	23.8	65.3	27.3	62.4	26.4	58.8	25.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.2	7.9	-10.4	37.6	4.0	0.8	-13.3	38.2	-15.6	38.6	-17.9	39.1	-20.8	39.7
			-11.9	21.1	3.5	0.8	-14.4	21.7	-16.5	22.1	-18.4	22.6	-21.0	23.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	1.2	3.1	9.7	14.8	19.7	24.2	26.7	25.7	21.9	15.7	9.0	4.0		
	DBStd	9.25	3.44	3.81	3.75	3.49	3.13	2.41	1.70	1.76	2.38	3.07	2.90	2.93		
	HDD10.0	762	273	194	51	4	0	0	0	0	0	1	52	187		
	HDD18.3	2184	532	426	268	114	23	1	0	0	3	93	280	444		
	CDD10.0	2481	0	2	42	147	301	427	516	488	357	178	22	1		
	CDD18.3	861	0	0	0	8	66	178	258	229	111	12	0	0		
	CDH23.3	10658	0	0	7	140	798	2241	3362	2704	1201	203	2	0		
	CDH26.7	4739	0	0	0	24	237	993	1732	1304	413	36	0	0		
(14) Wind	WSAvg	2.8	2.1	2.2	2.5	2.8	3.5	4.4	4.1	3.3	2.7	2.3	1.9	1.9		
(15) Precipitation	PrecAvg	388	43	70	67	51	32	16	16	22	16	14	20	21		
	PrecMax	657	102	139	167	122	130	58	47	88	47	58	76	70		
	PrecMin	124	5	5	5	8	3	2	2	1	0	0	0	2		
	PrecStd	121	28	36	40	30	27	14	10	18	14	17	19	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	18.9	24.1	29.2	32.2	35.9	37.1	36.2	33.2	29.8	23.1	17.1		
		MCWB	3.2	6.8	10.0	11.8	12.7	14.3	15.3	14.9	13.9	11.2	8.5	5.8		
	2%	DB	11.2	14.1	21.8	26.9	30.9	34.2	35.8	35.0	31.9	27.8	20.2	14.1		
		MCWB	2.5	5.2	9.0	11.5	12.4	13.7	15.3	14.6	13.4	11.1	7.1	4.3		
	5%	DB	9.2	11.8	19.2	24.9	29.2	33.0	34.8	33.8	30.2	25.8	18.1	12.1		
		MCWB	1.6	4.9	8.1	11.3	12.0	13.6	14.7	14.4	13.0	10.4	6.3	3.3		
	10%	DB	7.1	9.2	17.1	22.9	27.9	31.8	33.8	32.2	29.1	23.9	16.1	10.2		
		MCWB	1.0	3.6	7.5	10.7	11.6	13.2	14.5	14.3	12.7	9.7	5.8	2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	8.0	11.4	14.2	14.9	19.0	20.9	20.5	19.0	15.4	10.0	7.1		
		MCDB	8.8	13.2	19.4	23.6	25.2	26.7	28.9	27.4	26.4	22.8	16.8	15.0		
	2%	WB	3.5	6.8	10.4	13.0	13.9	17.0	19.9	19.7	17.8	13.5	8.8	5.6		
		MCDB	7.8	12.4	17.7	22.0	24.1	26.7	27.7	26.5	25.2	21.6	16.4	11.6		
	5%	WB	2.6	5.9	9.5	12.2	13.2	15.1	19.1	19.0	16.8	12.0	7.8	4.4		
		MCDB	6.8	9.9	16.2	21.2	23.9	27.8	26.9	25.8	24.1	20.7	14.9	9.6		
	10%	WB	1.7	4.7	8.7	11.4	12.7	14.0	18.0	18.2	15.6	10.8	6.9	3.4		
		MCDB	5.6	8.4	15.2	19.9	23.5	29.1	25.9	25.3	23.5	21.1	13.8	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	10.0	11.4	12.7	14.2	14.3	14.2	14.0	14.7	15.1	13.3	12.5		
		MCDBR	13.8	13.4	15.1	16.0	16.0	15.9	15.6	15.6	15.5	16.6	16.4	15.3		
	5% WB	MCWBR	7.3	6.7	5.9	5.4	4.9	5.4	6.0	6.2	5.9	6.2	7.3	7.7		
		MCWBR	10.9	11.1	12.1	13.1	13.7	14.2	13.5	12.7	13.1	13.8	13.3	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.327	0.377	0.410	0.415	0.430	0.440	0.411	0.357	0.336	0.306	0.285		
	taud		2.391	2.227	2.133	2.098	2.113	2.069	2.053	2.150	2.293	2.310	2.394	2.420		
	Ebn at Noon		928	911	890	876	872	855	844	864	900	891	891	904		
	Edh at Noon		92	121	146	159	158	165	167	149	123	110	91	84		
(44) All-Sky Solar Radiation	RadAvg		2.96	3.67	4.88	6.05	7.31	8.09	7.98	7.25	6.38	5.07	3.50	2.87		
	RadStd		0.28	0.43	0.46	0.47	0.45	0.35	0.22	0.40	0.24	0.23	0.27	0.35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page